

Year 8 Science – Booklet 4: Physics

Forces in Action — ANSWERS

Forces – Quick Test (p.91)

1. It may start moving (if stationary), stop (if already moving), speed up, slow down, change direction, or change shape.
2. To measure the size of a force.
3. Newtons (N).
4. No effect – the object's motion stays the same: if stationary it stays stationary, if moving it continues at the same speed and direction.
5. The size of the force applied (bigger force = greater acceleration) and the mass of the object (bigger mass = smaller acceleration for the same force).

Forces – Section A (5 marks)

1. c) it may travel at a constant speed
2. c) a car accelerating
3. a) weight
4. d) lift
5. d) mass

Forces – Section B (12 marks)

1. Match the words to their definitions (7 marks)

balanced → these forces will have no effect on the motion of an object

newton → this is the unit we use to measure forces

upthrust → a force exerted upon an object placed in a liquid

unbalanced → these forces will change the motion of an object

newtonmeter → an instrument for measuring the size of a force

weight → the gravitational force that pulls an object downwards

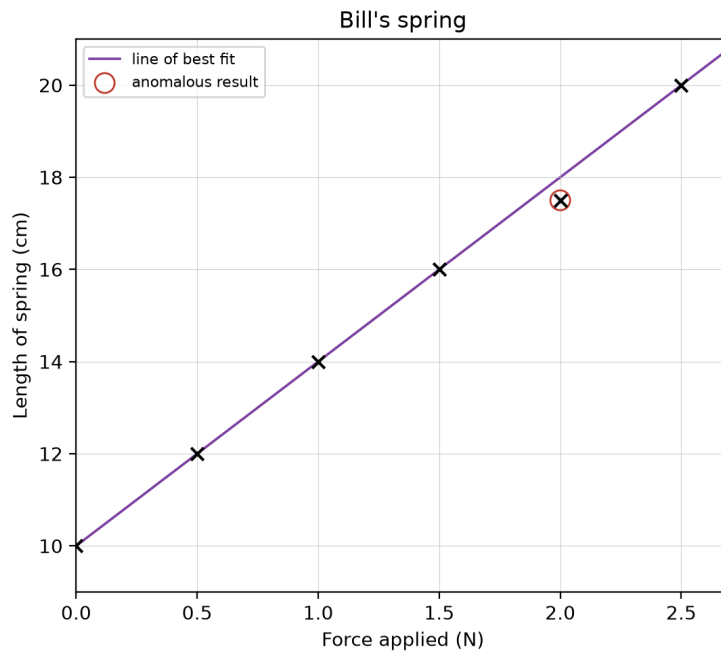
magnet → this object can apply forces to other objects without being in contact

2. Four skaters, diagrams A–D

- a). Largest acceleration: **D**
- b). Second largest acceleration: **A**
- c). Smallest acceleration: **C**
- d). Friction and air resistance.

Forces – Section C (12 marks)

1. Bill's newtonmeter spring experiment



c). The reading at 2.0 N (17.5 cm) doesn't fit the pattern – based on the trend (the spring extends 4 cm for every extra 1 N), the length at 2.0 N should be about 18.0 cm, not 17.5 cm.

d). Using the line of best fit (length = 10 + 4 × force): length = 10 + 4 × 0.75 = **13.0 cm**

e). 15.0 = 10 + 4 × force → force = 5 ÷ 4 = **1.25 N**

f). With the same 5 N weight now shared between two identical springs in parallel, each spring only carries half the force (2.5 N each instead of 5 N). Since extension is proportional to force, each spring stretches by half as much: 4.0 ÷ 2 = **2.0 cm**.

Friction and Terminal Velocity – Quick Test (p.93)

1. In the opposite direction to the object's motion (or attempted motion) – it opposes movement.
2. Shaping an object so it cuts through air (or water) with less resistance, reducing drag.
3. A substance (e.g. oil or wax) used to reduce friction between two surfaces in contact.
4. Any valid example, e.g. friction between tyres and the road lets a car grip, steer and brake safely; friction between shoes and the ground stops you slipping when walking.
5. It travels at a constant speed – its terminal velocity, if falling/moving through a fluid.
6. Heating (a rise in temperature) and wearing away (wear) of the surfaces in contact.

Friction and Terminal Velocity – Section A (5 marks)

1. b) temperature
2. b) roughening the surfaces
3. a) having a streamlined shape
4. a) increase
5. d) terminal velocity

Friction and Terminal Velocity – Section B (12 marks)

1. Fill in the missing words (10 marks)

At the start of a bobsleigh race the team has to push the bobsleigh as hard as they can. The team members wear **spikes** on the bottoms of their shoes to prevent them from slipping. As the sleigh gains

speed the team jump on board and crouch down so that they are **streamlined**. As the bobsleigh travels down the run it gains speed. The **frictional** forces between the sleigh and the **ice** and between the runners and the **ice** increase. To keep these forces to a minimum the runners are coated with a **lubricant** such as **wax** and the shape of the sleigh is **streamlined**.

2. Advantage/disadvantage of friction in sport

- a). Advantage: e.g. rock climbing – friction between hands/shoes and the rock lets you grip and avoid slipping.
- b). Disadvantage: e.g. speed skating/cycling – too much friction (surface or air resistance) slows the athlete down.

Friction and Terminal Velocity – Section C (9 marks)

1. Skydiver falling

- a). Gravity (her weight).
- b). It increases – she accelerates.
- c). Air resistance (drag).
- d). They are balanced (air resistance equals her weight), so there's no resultant force and she falls at a constant speed.
- e). Opening the parachute greatly increases her surface area, which greatly increases air resistance. This becomes much bigger than her weight, creating a net upward force that decelerates her until a new, lower terminal velocity is reached.

2. Bicycle brake blocks

- a). They get hot (from friction), and they wear away/wear down over time.
- b). The chain (also accept: the wheel axles/bearings, gears).